

Bhavesh Ghanchi

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EDUCATION

University of Southern California, Los Angeles, CA | MS Computer Science | GPA: 3.83/4.00 May 2026
K. J. Somaiya College of Engineering, Mumbai, India | BTech Computer Engineering | CGPA: 9.64/10 Jun 2024

EXPERIENCE

Easley-Dunn Productions

Torrance, CA

Software Engineer Intern | Co-Lead, Multiplayer Core Gameplay Team

Jul 2026 - Present

- **Architected** a reusable **C# vehicle-simulation and state-machine framework** that unified Robot, Car, and Plane handling while centralizing physics tuning, transformations, drifting, combat, and recovery across **6 tracks**.
- **Built a fault-tolerant multiplayer synchronization system** for **6-player races**, delivering **67% fewer transform updates through 10 Hz synchronization** against a **30 Hz server tick** while supporting connectivity recovery and data-driven AI balancing.

Streets Are For Everyone (SAFE)

Los Angeles, CA

Software Developer Intern

Jun 2025 - Aug 2025

- **Architected** a **geospatial intelligence platform** using **React, Node.js, and Python**, designing an ETL and graph-construction pipeline that converted **497 GIS features into 1,248 routable segments**, increasing routing granularity by **151%** with **100% coverage**.
- **Launched** an **LLM-assisted route planner** for a **60-cyclist beta**, extracting safety preferences from natural-language requests and grounding recommendations in **GIS data**, with rider feedback guiding improvements before wider rollout.

Colgate-Palmolive

Mumbai, India

Full-Stack Intern

Jan 2024 - Jun 2024

- **Built** a scalable **React and TypeScript design system** with **25+ reusable components**, standardizing the Hill's Pet Hub experience and accelerating consistent frontend delivery across **7 international markets**.
- **Automated multi-environment releases** through **CI/CD pipelines and deployment configuration**, supporting a production launch to **10,000+ French users** and establishing a repeatable rollout process for **6 additional countries within 3 months**.
- **Reduced peak API query latency by 25%** by optimizing **SQL queries** and introducing **Redis caching** for high-traffic workloads.

K. J. Somaiya College of Engineering

Mumbai, India

Research Intern

Mar 2023 - May 2023

- **Built a Python clinical NLP pipeline** processing **5,000+ healthcare documents**, fine-tuning **BERT with SciSpacy** to extract medical entities and classify content across **12 categories**, achieving an **89% F1 score**.

SayHey

Mumbai, India

Backend Developer Intern

Jun 2022 - Aug 2022

- **Owned the backend** for a mental-health platform, developing APIs, booking workflows, availability validation, and conflict prevention that **reduced latency by 30%** and operating costs by **~15%** after replacing third-party scheduling software.

PROJECTS

OTAS: Observability Tool for Agentic AI Systems | github.com/bhaveshGhanchi/otas

Apr 2026

Python, Django, PostgreSQL, Redis, Celery, OpenTelemetry, MCP, Docker | Team project + personal fork

- **Co-engineered an observability platform for agentic AI systems**, capturing prompts, tool calls, latency, and outcomes while reconstructing execution lineage across multi-agent runs, handoffs, retries, and failures.
- **Built OpenTelemetry tracing and a Pydantic-validated MCP ingestion server** with CLI hooks for Claude Code, Cursor, and Codex; load-tested **10 concurrent agents across 800 sessions**, persisting **1,620 events at 149 ms p95 latency**.

LEAP: Loss-Aware End-to-End Acknowledged Protocol | github.com/bhaveshGhanchi/leap

May 2026

Java, UDP Sockets, Maven

- **Implemented a TCP-style reliable transport protocol from scratch over UDP**, including sliding windows, cumulative ACKs, fast retransmission, adaptive RTO, Tahoe congestion control, and CRC32 with end-to-end SHA-256 integrity checks.
- **Built a benchmarking harness with 3 packet-loss simulation modes**, achieving **46.2 MB/s throughput at 0% loss** and maintaining **88.8% transfer efficiency at 10% loss**.

PUBLICATIONS

M. Garg, L. Wang, **B. Ghanchi**, S. Dumpala, S. Kakde, Y. Chen. "Biomedical Literature Q&A System Using RAG." *arXiv preprint arXiv:2509.05505*, Sep 2025.

- **Co-developed a biomedical RAG question-answering system** using **MiniLM embeddings, top-5 FAISS retrieval, and QLoRA-tuned Mistral-7B** on MedQuAD, achieving a **0.88–0.90 BERTScore F1** on breast-cancer literature.

SKILLS

Languages: Python, Java, C#, TypeScript, JavaScript, SQL

AI & ML: LLMs, RAG, Transformers, BERT, Embeddings, FAISS, QLoRA, Hugging Face, LLM Evaluation, Agentic Workflows, OpenAI API, MCP

Backend & Data: FastAPI, Django, Node.js, Express, REST APIs, Event-Driven Microservices, PostgreSQL, Redis, Celery, Firebase, React

Infrastructure & Observability: Docker, CI/CD, Linux, Git, OpenTelemetry